

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)

Total Marks: 30

Question Count: 3

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.

Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability.

Security is enforced through authentication tokens, HTTPS, and role-based access control.

The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

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1. Case Study 1: Smart Retail Inventory Management System

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Answer:

1. Introduction

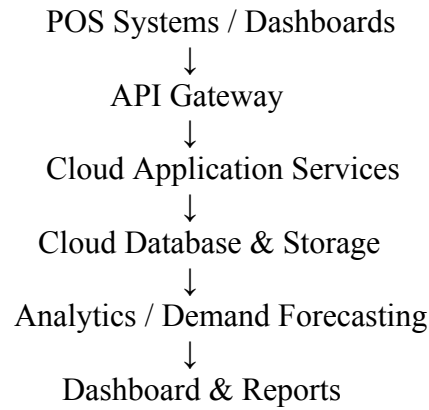
The Smart Retail Inventory Management System is a cloud-based solution designed to manage product inventory across multiple retail stores. It connects POS systems, dashboards, cloud storage, APIs, and analytics services through a centralized cloud platform. The system helps retailers monitor stock levels, analyze sales, and forecast future product demand.

2. Cloud Architecture

The system consists of the following major components:

- **Clients:** POS terminals and web dashboards used by store employees and managers.
- **API Layer:** APIs allow clients to securely communicate with cloud services.
- **Cloud Services:** Application and platform services process inventory and transaction data.
- **Cloud Storage:** Stores product information, transaction logs, and analytics datasets.
- **Analytics Platform:** Processes historical sales data and performs demand forecasting.
- **Security Layer:** VPNs, firewalls, authentication, and API gateways protect the system.
- **Web Applications:** Provide inventory dashboards, reports, and management functions.

3. Working Flow



When a product is sold through a POS system, the transaction is sent through the API gateway to the centralized cloud service. The inventory database is updated automatically. The transaction is also stored for analytics. The analytics platform can analyze historical sales and predict future demand, helping the retailer maintain appropriate stock levels.

4. Cloud Service Models

Cloud Model	Application in the System
IaaS	Virtual machines, networking, storage and computing infrastructure
PaaS	Analytics, databases, APIs and demand forecasting services
SaaS	Inventory dashboards and web-based management applications

5. Security Mechanisms

The system uses multiple security mechanisms:

- **VPN** – Provides secure communication between stores and cloud infrastructure.
- **Firewall** – Controls and filters network traffic.
- **API Gateway** – Controls and secures API requests.
- **Authentication & Authorization** – Ensures only authorized users access the system.
- **Data Encryption** – Protects sensitive data during transmission and storage.
- **Access Control** – Provides different permissions to administrators, managers, and employees.

6. Benefits

- Real-time inventory monitoring
- Centralized data management
- Reduced manual inventory work
- Improved demand forecasting
- Easy access from multiple stores
- Scalable cloud infrastructure
- Better decision-making using analytics
- Improved security and data availability

7. Conclusion

The Smart Retail Inventory Management System demonstrates how IaaS, PaaS, and SaaS can be integrated into a single cloud-based solution. POS systems and dashboards communicate with centralized cloud services through APIs, while cloud storage and analytics services manage and analyze inventory data. Security mechanisms such as VPNs, firewalls, and API gateways ensure secure operation. Thus, the proposed system provides a scalable, secure, and efficient solution for modern retail inventory management.

Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

1. Introduction

The Cloud-Based Document Collaboration System is a cloud application similar to Google Docs, which allows multiple users to create, edit, share, and manage documents through web browsers. Documents are stored in distributed cloud storage, enabling users to access them from different devices and locations.

2. Cloud Architecture

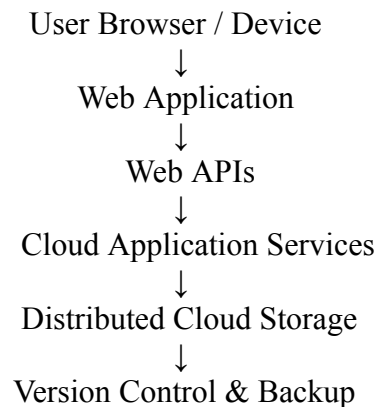
The system consists of the following major components:

- Clients: Web browsers, laptops, tablets, and mobile devices used by users.
- Web Application: Provides the document editing and collaboration interface.
- Web APIs: Synchronize document changes between users and devices.
- Cloud Storage: Stores documents and related files in distributed storage.
- Version Control: Maintains previous versions of documents and allows users to

recover earlier versions.

- **Network Infrastructure:** Provides low-latency communication and high availability across different geographical locations.
- **Security Layer:** Provides encryption, identity management, authentication, and access permissions.

3. Working Flow



When a user edits a document, the changes are sent to the cloud through web APIs. The cloud service synchronizes the changes with other connected users in real time. The updated document is stored in distributed cloud storage, while version control maintains previous versions.

4. Cloud Service Models

Cloud Model	Application in the System
IaaS	Virtual servers, networking, storage, and computing resources
PaaS	Cloud databases, APIs, synchronization, and application development services
SaaS	Browser-based document editing and collaboration application

5. Security Mechanisms

- **Encryption:** Protects documents during transmission and storage.
- **Identity Management:** Identifies and manages registered users.
- **Authentication:** Ensures only valid users can access the system.
- **Access Permissions:** Controls who can view, edit, comment, or share documents.
- **Secure APIs:** Protect communication between clients and cloud services.
- **Version Control:** Maintains document history and helps recover previous versions.

6. Benefits

- Real-time collaboration between multiple users
- Access documents from anywhere
- Automatic cloud storage and synchronization
- Easy document sharing
- Version history and recovery
- High availability
- Reduced dependency on local storage
- Scalable infrastructure for large numbers of users

7. Conclusion

The Cloud-Based Document Collaboration System demonstrates how cloud computing enables real-time document editing, sharing, synchronization, and storage. Distributed cloud storage provides availability and reliability, while web APIs synchronize changes across multiple devices. Encryption, identity management, and access permissions provide security. Thus, the system provides a scalable, reliable, secure, and collaborative cloud-based document management solution.

Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

1. Introduction

The Online Food Ordering Platform is a cloud-based application that allows customers to browse menus, place food orders, make payments, and track orders using web browsers or mobile devices. The frontend communicates with backend cloud services through RESTful Web APIs. Cloud storage securely stores menu images, invoices, and customer information.

2. Cloud Architecture

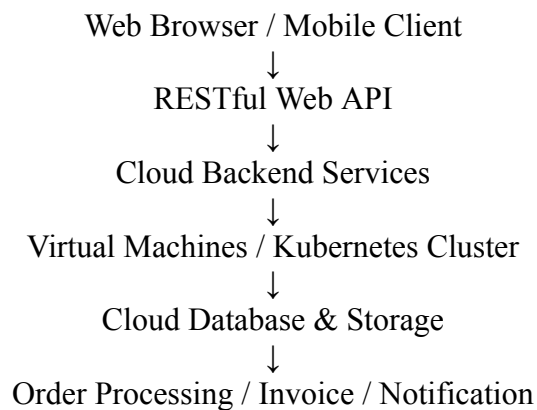
The system consists of the following components:

- Clients: Web browsers and mobile applications used by customers.
- Frontend: Provides the user interface for browsing menus and placing orders.
- RESTful Web APIs: Connect frontend clients with backend services.
- Backend Services: Process orders, customer details, payments, and restaurant

information.

- Cloud Storage: Stores menu images, invoices, and customer data.
- Virtual Machines: Provide computing resources for application services.
- Managed Kubernetes: Manages containers and automatically scales services based on demand.
- Security Layer: Uses authentication tokens, HTTPS, and role-based access control.

3. Working Flow



When a customer opens the application, the frontend requests menu information from the backend through RESTful APIs. The customer selects food items and places an order. The API sends the order to the backend service, which processes it and stores the required information in the cloud. Menu images and invoices are maintained in cloud storage. Kubernetes manages the application containers and provides scalability when the number of users increases.

4. Cloud Service Models

Cloud Model	Application in the System
IaaS	Virtual machines, networking, and computing infrastructure
PaaS	Managed Kubernetes, databases, APIs, and application platform services
SaaS	Web and mobile-based food ordering application

5. Security Mechanisms

- Authentication Tokens: Verify the identity of users and API clients.
- HTTPS: Encrypts communication between clients and cloud services.
- Role-Based Access Control (RBAC): Provides different permissions to customers, restaurant staff, and administrators.
- Secure Cloud Storage: Protects customer data, invoices, and other stored information.
- API Security: Controls and validates requests sent to backend services.

6. Benefits

- Online food ordering from anywhere
- Real-time order processing
- Scalable infrastructure during high traffic
- Secure storage of customer and invoice data
- Faster communication through RESTful APIs
- High availability and reliability
- Easy integration with web and mobile applications
- Efficient management using Kubernetes

7. Conclusion

The Online Food Ordering Platform demonstrates the integration of cloud clients, RESTful APIs, cloud storage, virtual machines, and managed Kubernetes in a real-time application. Security mechanisms such as authentication tokens, HTTPS, and role-based access control protect users and data. The use of cloud infrastructure and Kubernetes provides scalability, reliability, security, and efficient real-time food ordering services.